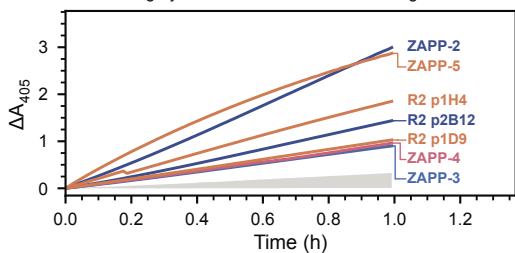


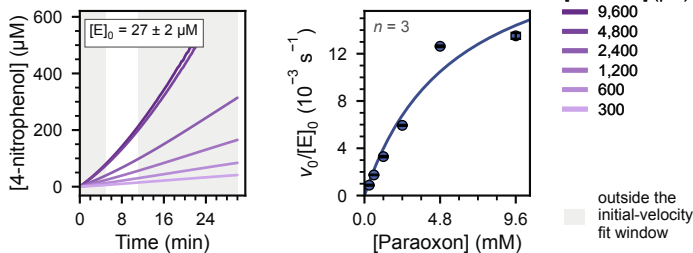
A Round 2 eluate screen ($n = 192$ designs)

600 μM paraoxon · 1% MeOH · 25 mM NaHCO_3 (final) · 25 °C, 1 h
gray = all other uncharacterized designs



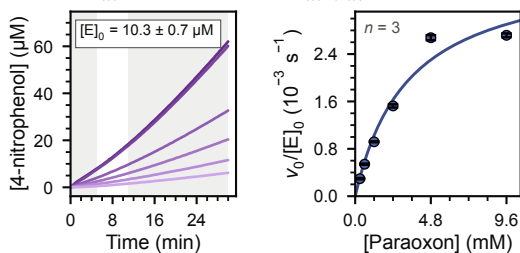
B ZAPP-2 (R2 p2C4)

$k_{\text{cat}} = 0.023 \pm 0.006 \text{ s}^{-1}$ $K_M = 6 \pm 3 \text{ mM}$
 $k_{\text{cat}}/K_M = 4 \pm 3 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 500,000 \pm 200,000$



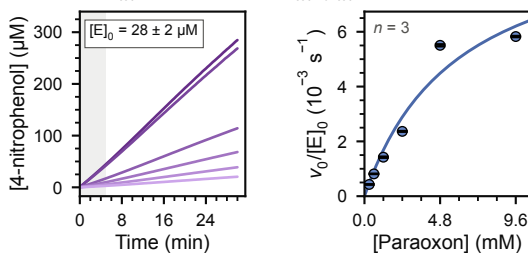
C R2 p2B12 (ZAPP-2 scaffold)

$k_{\text{cat}} = 0.0039 \pm 0.0006 \text{ s}^{-1}$ $K_M = 3 \pm 2 \text{ mM}$
 $k_{\text{cat}}/K_M = 1.2 \pm 0.5 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 80,000 \pm 30,000$



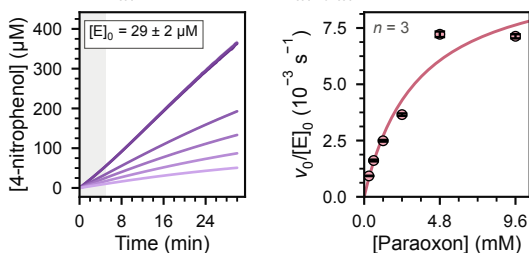
D ZAPP-3 (R2 p2G1)

$k_{\text{cat}} = 0.010 \pm 0.003 \text{ s}^{-1}$ $K_M = 6 \pm 4 \text{ mM}$
 $k_{\text{cat}}/K_M = 2 \pm 1 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 200,000 \pm 80,000$



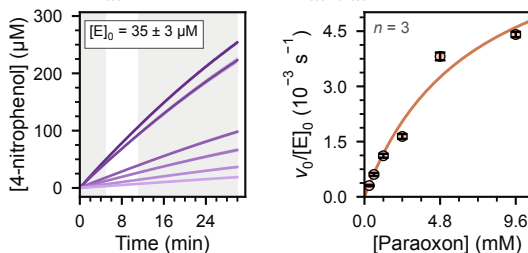
E ZAPP-4 (R2 p2E3)

$k_{\text{cat}} = 0.010 \pm 0.002 \text{ s}^{-1}$ $K_M = 3 \pm 2 \text{ mM}$
 $k_{\text{cat}}/K_M = 3 \pm 2 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 210,000 \pm 70,000$



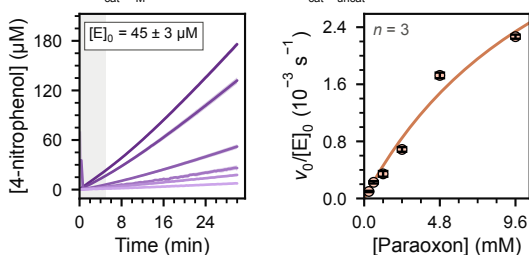
F ZAPP-5 (R2 p1F6)

$k_{\text{cat}} = 0.008 \pm 0.002 \text{ s}^{-1}$ $K_M = 7 \pm 3 \text{ mM}$
 $k_{\text{cat}}/K_M = 1.2 \pm 0.6 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 160,000 \pm 60,000$



G R2 p1D9 (ZAPP-5 scaffold)

$k_{\text{cat}} = 0.006 \pm 0.003 \text{ s}^{-1}$ $K_M = 14 \pm 8 \text{ mM}$
 $k_{\text{cat}}/K_M = 0.4 \pm 0.3 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 110,000 \pm 60,000$



H R2 p1H4 (ZAPP-5 scaffold)

$k_{\text{cat}} = 0.0025 \pm 0.0006 \text{ s}^{-1}$ $K_M = 7 \pm 3 \text{ mM}$
 $k_{\text{cat}}/K_M = 0.3 \pm 0.2 \text{ M}^{-1} \text{ s}^{-1}$ $k_{\text{cat}}/k_{\text{uncat}} = 50,000 \pm 20,000$

